

Bhavana Mehta

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EDUCATION

University of Pennsylvania

Ph.D. & M.S. in Computer and Information Science

2019 – 2025

Nirma Institute of Technology

B.Tech., Electronics & Communication Engineering

2014 – 2018

SELECTED PUBLICATIONS

ZAYA1-8B Technical Report (*arXiv*, May 2026)

Bhavana Mehta et al. (in collaboration with Zyphra)

Adaptive Sharding in Untrusted Environments (*SIGMOD' 26*)

Bhavana Mehta, Nupur Baghel, Mohammad J Amiri, Ryan Marcus, Boon Thau Loo

Towards Full Stack Adaptivity in Permissioned Blockchains (*VLDB '24*)

Chenyuan Wu, Mohammad J Amiri, Haoyun Qin, **Bhavana Mehta**, Ryan Marcus, Boon Thau Loo

Towards Adaptive Fault-Tolerant Sharded Databases (*VLDB '23*)

Bhavana Mehta, Neelesh C A, Prashanth Iyer, Mohammad J Amiri, Boon Thau Loo, Ryan Marcus

AdaChain: A Learned Adaptive Blockchain (*VLDB '23*)

Chenyuan Wu, **Bhavana Mehta**, Mohammad J Amiri, Ryan Marcus, Boon Thau Loo

Human Action Recognition using Good Features and Multilayer Perceptron Network (*ICCS' 17*)

J. Talukdar, **Bhavana Mehta**

EXPERIENCE

Zyphra

Member of Technical Staff, Performance Engineering

Dec 2025 – Present

- **ML Training Control Plane:** Built and operated Zookeeper, Zyphra's internal control plane for large-scale LLM training on AMD GPU clusters, enabling Slack-native job controls, automated rekicks, checkpoint sync management, node exclusion workflows, and real-time operator intervention during active training runs.
- **Distributed Checkpointing & Training Recovery:** Designed recovery systems for failure-prone multi-node training, including pairwise NVMe checkpoint replication, failed-node replacement workflows, and object-store checkpoint repair; reduced operational risk across live LLM training runs by hardening recovery paths for checkpoint durability and restart continuity.
- **GPU Systems Reliability & Root-Cause Analysis:** Diagnosed and resolved reliability-critical failures across AMD GPU training environments, including ROCm/HIP memory anomalies, Prometheus storage incidents, monitoring regressions, and cluster-level storage failures; translated low-level system behavior into actionable fixes for active large-scale workloads.
- **MI355X Inference Optimization & Benchmarking:** Contributed to Zyphra's AMD MI355X inference benchmarking and full-stack optimization effort, characterizing TTFT, output throughput, kernel behavior, RCCL/HIP stack performance, communication patterns, and high-throughput serving metrics across large-model inference workloads.

University of Pennsylvania

Research Fellow, Distributed Systems Lab (Advisor: Prof. Boon Thau Loo)

2019 – 2025

- **AdaChain (RL for Systems):** Built a **reinforcement learning**/contextual-bandit framework for adaptive blockchain execution, including ledger state featurization, online model training, and decentralized architecture switching.
- **Marlin (Algorithmic Sharding):** Led adaptive sharding for untrusted distributed systems; benchmarked KaHyPar-based centralized resharding and decentralized consensus protocols, improving throughput by **over 40%** under adversarial skew.
- **Fault-Tolerant Systems:** Designed health monitoring/state replication for 70B+ LLM serving failover and built a DPDK TCP migration prototype with Microsoft Research/MIT, achieving $\sim 4\mu\text{s}$ migration latency and a **12x improvement**.

Bluespec Inc.

Design Engineer (prev. Intern)

2018 – 2019

- Tuned RISC-V CPU pipeline and **branch predictor (statistical modeling)** for compute-intensive workloads, **improving IPC by 25%**; concurrently built a Jenkins/Docker-based CI/CD pipeline that reduced hardware release cycles from two weeks to three days.

IIT Madras, RISE Lab

Research Intern

Summer 2017

- Designed a novel SRT Radix 4 algorithm for high-speed division; this design was integrated into India's first **open-source RISC-V processor** (SHAKTI), improving parallel quotient selection.

SKILLS

Languages:

Python, C++/C, SQL, Bash

Machine Learning:

Reinforcement Learning, Contextual Bandits, PyTorch, Scikit-learn

Distributed Systems:

Raft/Paxos, PBFT/HotStuff, 2PC, Redis, KaHyPar, YCSB

ML Infrastructure:

Distributed training orchestration, Checkpoint validation & recovery, Slurm, NFS/NVMe, Prometheus, Backblaze B2, vLLM, Perferetto

Systems & DevOps:

Linux, Docker, Git, Jenkins, CI/CD, HPC, CloudLab, Performance Optimization

Networking:

DPDK, Kernel Bypass, Low-Latency Networking